Harvesting under a strong Allee effect

Collapse thresholds and PINN analysis

RaghavSinghal15, AshChan07, devam2006
Prem-Mehta03, rohanagrawal28

Working draft, 28 October 2026

Models and scaling

Constant catch

$$A: x' = x(1 - x)(x/a - 1) - q$$

Proportional effort

$$B: x' = x(1 - x)(x/a - 1) - ex$$

$x = N/K \quad \tau = rt \quad a = A/K$
$q = H/(rK) \quad e = h/r$

One deterministic population, constant parameters, $0 < A < K$

Equilibria and harvesting history

WP01: foundations and Model A

Awaiting the corrected equilibrium, stability
and branch-diagram contribution

WP02: comparison and harvesting history

Awaiting matched-policy comparisons, harvest ramps
and explicitly labelled restocking experiments

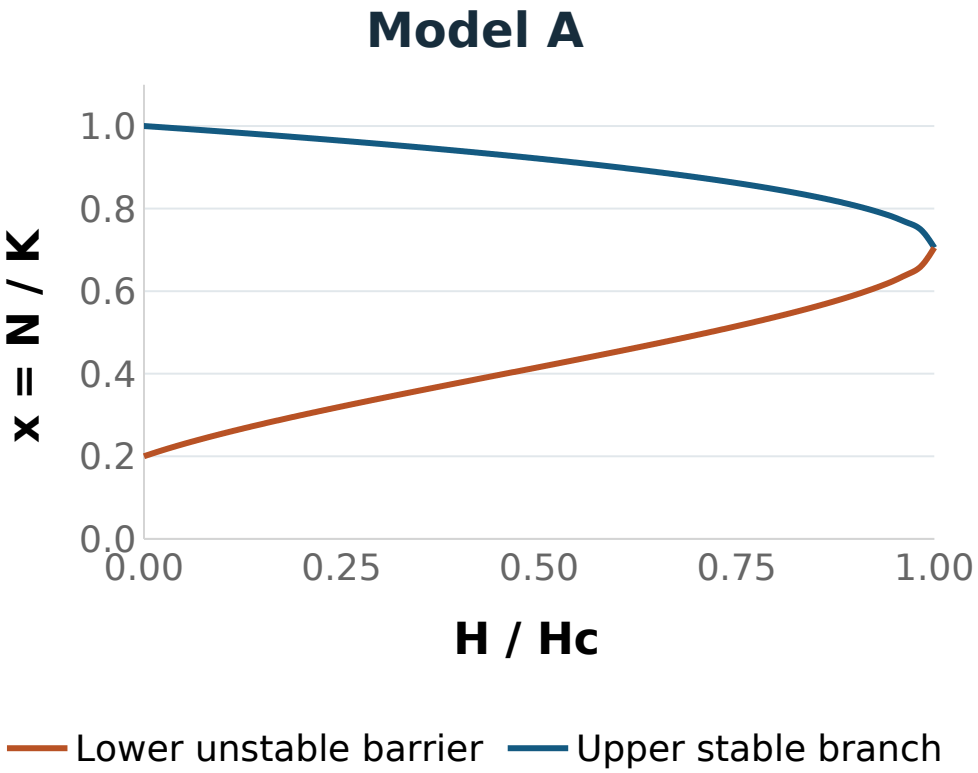
Harvesting thresholds

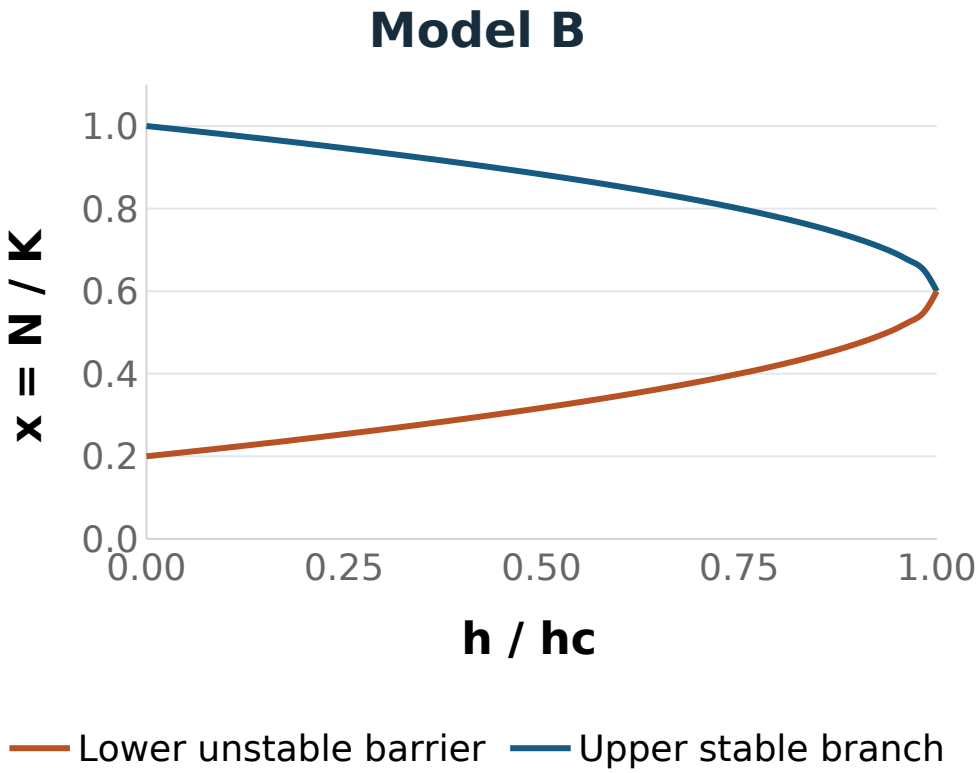
Illustrative parameters: r = 0.7, K = 100, A = 20

Quantity	Value	Units
A: N*	70.55050	Population
A: Hc	36.75966	Population / time
B: hc	0.56	1 / time
Logistic: Hc	17.5	Population / time

A maximises total growth. B maximises growth per individual.
A population barrier still matters below either threshold.

Initial population separates the outcomes





Below the lower branch: collapse. Above it: approach the upper branch.
At the fold, attraction occurs only from above.

Working draft, synthetic examples

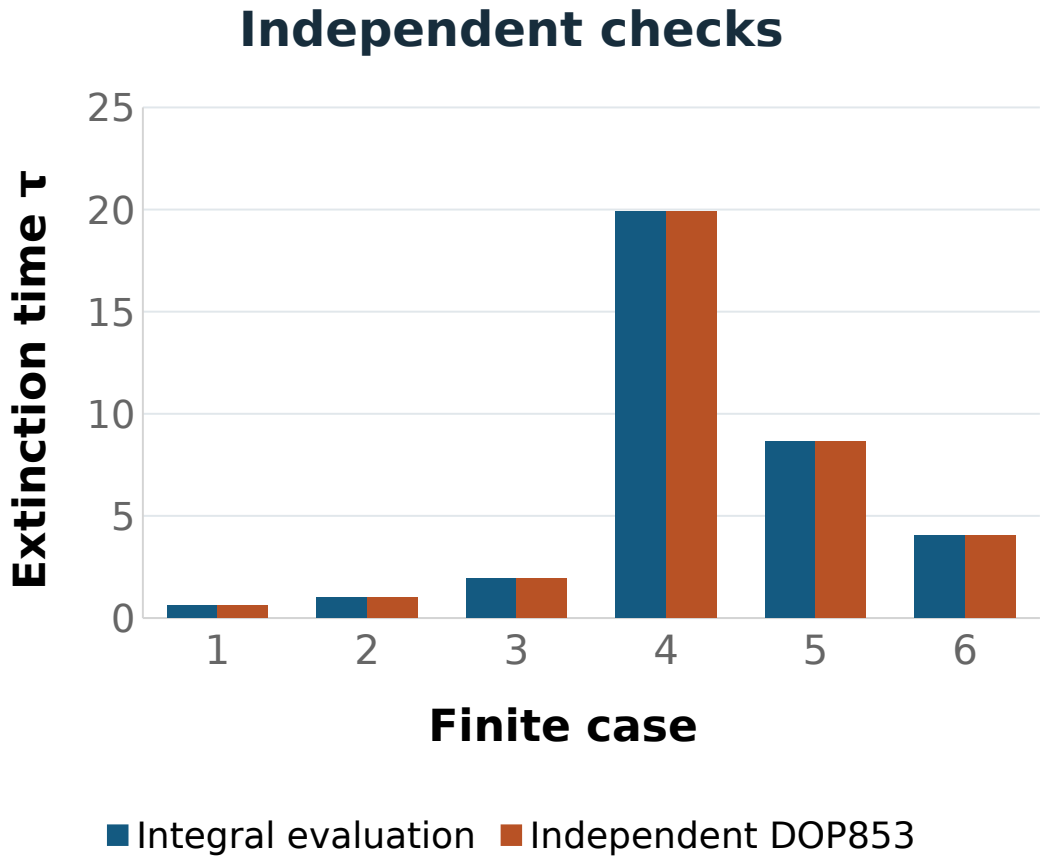
A reaches zero in finite time

$\tau_{ext} = \int [q - G(u)]^{-1}\, du$ from 0 to x_0

Only reaching-zero paths enter the integral.

RK45 stops at the first downward zero crossing.

B extinction paths approach zero asymptotically.



Forward and inverse PINN evidence

WP04: forward problem

Awaiting actual A/B predictions,
RMSE and relative error,
runtime and near-collapse failure

WP05: inverse problem

Awaiting parameter recovery,
noise and held-out validation,
and an ODE-fitting baseline

No numerical PINN result is included in this draft.

Classical findings and limits

Initial population matters even below the harvesting cliff.

Fixed catch can cause finite-time extinction.
On extinction paths, B approaches zero asymptotically.

A long bottleneck can hide an eventual collapse.

These conclusions concern the assumed deterministic model.
All current examples are synthetic. PINN validation is pending.

Questions

Thresholds, basins, extinction and numerical checks

Final allocation: 25-minute talk and 5-minute Q&A